

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Prove mole% = pressure % = volume%

Q.24 The feed contains 50% benzene and 50% toluene is fed to a distillation column at the rate of 5000 kg/h. A top product contains 95% benzene and the bottom product contains 92% toluene. All percentages are by weight. Calculate

- i) the mass flow rate of top and bottom products
- ii) Percentage recovery of benzene

Q.25 Write short note on any two of the following:

- a) combustion
- b) heat of combustion
- c) heat of formation
- d) heat of reaction

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3rd Sem / Chemical, Chemical (Pulp, & Paper), Chem Engg. (Spl. Paint Tech.) Chem Engg. (Spl. Polymer Engg.)

**Subject : Chemical Process Calculations /
Ind. Chem Cal.**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Concept of Energy balance is based upon

- a) Conservation of mass
- b) Conservation of energy
- c) Conservation of volume
- d) None

Q.2 Full form of CGS system is

- a) Central gross system
- b) Centimeter gross system
- c) Centimeter Gram Second
- d) None

Q.3 Relation between C_p and C_v is

- a) $C_p - C_v = 0$ b) $C_p + C_v = 0$
c) $C_p = C_v$ d) $C_p - C_v = R$

Q.4 10 dm is expressed as

- a) 100 mm b) 100 cm
c) 100 m d) 100 km

Q.5 $10^\circ\text{C} = \underline{\hspace{2cm}}\text{K}$

- a) 10 b) 283.15
c) 373.15 d) 0

Q.6 C_p is the heat capacity at

- a) Common platform b) constant parameters
c) Constant pressure d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 One pound has _____ grams

Q.8 One mm Hg= _____ Torr.

Q.9 Define mole.

Q.10 Write the molecular weight of H_2O

Q.11 Material balance is based on law of conservation of _____

Q.12 Define Molality.

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SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write the basic steps follows for a material balance calculation.

Q.14 90 gm of NaOH are dissolved in water to prepare 500 ml solution. Find the normality and molarity of the solution.

Q.15 With the help of neat diagram explain the purge stream of material balance.

Q.16 Explain the terms Molarity and Normality with their Mathematical relations.

Q.17 Define combustion, Sensible heat and latent heat

Q.18 What is a recycle stream? Describe with the help of diagram.

Q.19 A natural gas has the following composition by volume $\text{CH}_4=82\%$, $\text{C}_2\text{H}_6=12\%$, $\text{N}_2=6\%$. Calculate the composition by weight.

Q.20 How many moles of H_2SO_4 will contain 64 kg of sulphur

Q.21 Derive the relation between C_p and C_v

Q.22 Discuss the Hess law of constant heat summation.

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